



S. No.



AIR UNIVERSITY, ISLAMABAD

ANSWER BOOK: FINAL SEMESTER EXAMINATION

REGISTRATION ID. 15862 CLASS & SEC _____ SUBJECT AI

S.NO. OF EXTRA SHEET/S (IF USED) 1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____ 7. _____

EXAMINATION (Please ☒ the appropriate Box) ☒ FALL ☐ SPRING ☐ SUMMER 20 25SIGNATURE OF STUDENT _____ DATE 9th Feb - 2020SIGNATURE OF INVIGILATOR _____ NAME OF INVIGILATOR ALMAS AYUB ANSARI

IMPORTANT INSTRUCTIONS

BEFORE STARTING YOUR WORK, ENSURE THAT:

1. YOU ARE IN POSSESSION OF YOUR PRINTED ROLL NUMBER SLIP AND AU ID CARD.
2. YOU ARE SITTING IN THE ROOM AND SEAT ASSIGNED TO YOU AS PER THE ROLL NUMBER SLIP.
3. YOU DO NOT HAVE ANY UNAUTHORISED PRINTED OR WRITTEN MATERIAL, CELL PHONES, SMART DEVICE OR ANY OTHER MATERIAL THAT MAY BE AN UNFAIR AID AS PER AU ACADEMIC REGULATIONS. DISCOVERY OF ANY SUCH MATERIAL AFTER START OF THE EXAMINATION WILL RENDER YOU LIABLE FOR DISCIPLINARY ACTIONS UNDER CHAPTER XV OF AU REGULATIONS.
4. WRITE YOUR REGISTRATION ID AND PARTICULARS CAREFULLY ONLY IN THE SPACES SPECIFIED ABOVE FOR THIS PURPOSE.

DURING THE EXAMINATION:

5. ENSURE THAT ANSWER BOOK AND ALL CONTINUATION SHEETS ARE SIGNED BY THE INVIGILATOR.
6. ANSWERBOOK CONTAINS 14 PAGES. TEARING OF ANY PAGE FROM THE ANSWERBOOK OR CONTINUATION SHEET IS AN OFFENCE UNDER AU UNFAIR MEANS POLICY.
7. DO NOT WRITE ANYTHING ON QUESTION PAPER EXCEPT YOUR REG ID. USE LAST PAGES OF ANSWERBOOK FOR ROUGH WORK.
8. DO NOT BORROW ANYTHING FROM OTHERS. RAISE YOUR HAND TO SEEK ASSISTANCE OF THE INVIGILATOR.
9. DO NOT USE LEAD PENCIL (EXCEPT FOR DIAGRAMS/GRAPHS ETC.) OR ANY INK OTHER THAN BLUE OR BLACK.
10. WRITE ON BOTH SIDES OF EACH PAGE IN THE ANSWERBOOK.
11. ENSURE THAT SERIAL NUMBERS OF ANSWERBOOK AND CONTINUATION SHEETS ARE ENTERED ON ATTENDANCE SHEET. ALSO ENTER SERIAL NUMBER OF ALL CONTINUATION SHEETS IN THE SPACE PROVIDED ON THE ANSWERBOOK.

AFTER THE EXAMINATION:

12. MAKE SURE THAT ALL CONTINUATION SHEETS ARE STAPLED WITH MAIN ANSWERBOOK AT THREE PLACES BEFORE SUBMITTING YOUR ANSWER SCRIPT TO THE INVIGILATOR.
13. DO NOT LEAVE EXAM HALL WITHOUT PERMISSION OF INVIGILATOR EVEN IF YOU HAVE COMPLETED THE EXAM.
14. IN NO CASE SHOULD YOU LEAVE THE EXAM HALL BEFORE HALF TIME OR YOUR PAPER MAY BE CANCELLED.

Q. NO.	1	2	3	4	5	6	7	8	9	10	TOTAL
MAXIMUM MARKS	20	20	30	30							100
MARKS OBTAINED	4	18	12	8							42

MARKS OBTAINED IN WORDS

Forty Two

EXAMINER'S SIGNATURE

HP

(2)

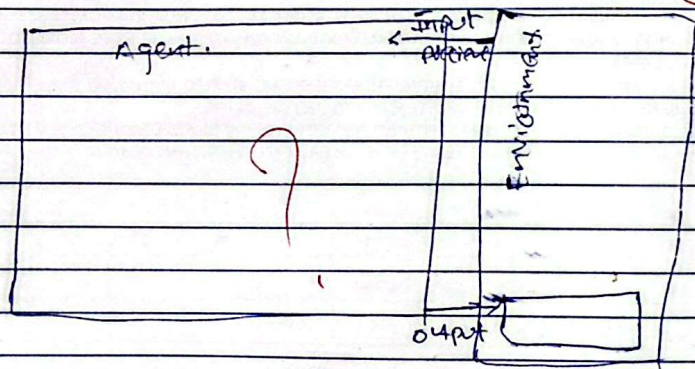
"(Question:-01)"

(A) Artificial intelligence:-

Artificial intelligence provides correctness, completeness, sustainability to any task and ensure that it's complete, on time and correct.

b) Agent:-

Agent in AI is perceives it data from environment.

c)-

(3)

d) Environment:-

Anything outside the Agent affect its performance is called environment.

Example:-

Box (wind, Road, weather are its Environment).

Type of Environment:-

- 1) Static
- 2) Dynamic (Ludo game)
- 3) Continuous (snake game)
- 4) Discrete (Tic tac toe)

Define ?

Question:-02

A) Informed search:-

Informed search in Artificial intelligence uses heuristic functions which helps to search the goal easy by the help of heuristic values.

b) Greedy Best-First Search:

Greedy Best-First first work with the heuristic value to find the best path to our desired goal.

Advantage:

It uses heuristic function.

Disadvantage:

Not optimal

Not complete.

c) A* search algorithm.

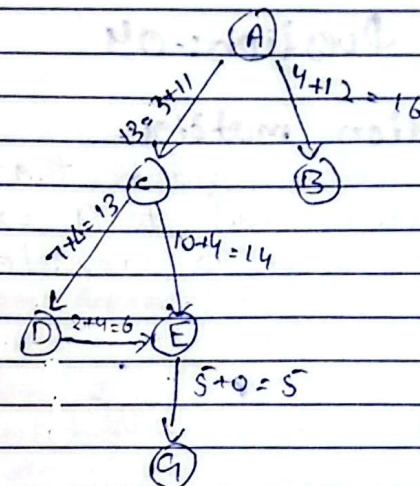
A* search uses with path cost and node cost to get to the desired goal.

$$f(n) = g(n) + h(n)$$

$\downarrow$ $\downarrow$
path cost

And it also uses heuristic information.

d)



Path = A $\rightarrow$ C $\rightarrow$ D $\rightarrow$ E $\rightarrow$ G.

$$h(n) = 14 + 13 + 6 + 4 + 0 = 35.$$

$$g(n) = 3 + 7 + 2 + 5 = 17.$$

$$f(n) = g(n) + h(n)$$
$$= 52$$

6

Question:-04

A. Confusion matrix:-

Confusion matrix predicts the classification model. It uses the data set values to evaluate.

9

b)-

TP	FN
FP	TN
50	10
5	35

so for accuracy:-

$$\frac{TN + TP}{\text{Total}} = \frac{85}{100} = 0.85$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} = \frac{50}{50 + 10} = \frac{50}{60} = \frac{5}{6}$$

$$= 0.833$$

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$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{35}{35 + 5} = \frac{35}{40} = \frac{7}{8}$$

$$= 0.875$$

C) Bayesian Network:-

- 1)- It is certain.
- 2)- It uses probability to solve and uncertainty.
- 3)- It uses Acyclic graph to show the dependencies.

d)-

for The both True conditions:-

Probability of rain = 0.3

Sprinkler probability = 0.01

The probability of grass being wet

$$= 0.3 \times 0.01 \times 0.95$$

$$P(w) = 2.85 \times 10^{-3}$$

2) T | F | 0.80

$$P(w) = 0.3 \times 0.40 \times 0.80 = 0.096$$

3) - F | T | 0.70

probability of rain = 0.3.

if it doesn't rain = $1 - 0.3 = 0.7$

$$= 0.7 \times 0.01 \times 0.70$$

$$P(w) = 4.9 \times 10^{-3}$$

4) - F | F | 0.05

$$P(w) = 0.7 \times 0.40 \times 0.05$$

$$P(w) = 0.014$$

"(Question: 03)"

A) Define Machine Learning:-

Machine Learning is used to communicate with Computer because they don't understand english etc. In Machine Learning communication is based on binary 0's and 1's. Traditional programming used system that can be understood by humans but machine learning use made to be understood by computer.

B) - Supervised and unsupervised learning.

Supervised learning mean that it is supervised not unsupervised and it's used to classify and regression.

C) - K-Nearest Neighbor (KNN)

K-Nearest Neighbor is a supervised Machine Learning algorithm which is used for classification and regression.

It simplify by the data points between them.

k = is the Neighbour.

Nearest = data point's between.

Neighbour =

Classify k = neighbour.

if the data point matching is less than any other.

then it will be the neighbour.

d)

new data point = $P(5, 6)$.

$$\text{Point AP} = \sqrt{(5-2)^2 + (6-4)^2}$$

$$= \sqrt{13} = 3.60 \quad \checkmark$$

(12)

$$\text{BP} = \sqrt{(5-4)^2 + (6-6)^2}$$

$$= 1 \quad \checkmark$$

$$\text{CP} = \sqrt{(5-6)^2 + (6-8)^2}$$

$$= \sqrt{25 - 36 + (-28)}$$

$$= \sqrt{-11 - 28}$$

$$= \sqrt{5} = 2.23 \quad \checkmark$$

$$= 6.244$$

$$\text{DP} = \sqrt{(5-8)^2 + (6-10)^2}$$

$$= \sqrt{103}$$

$$= 5 \quad \checkmark$$

AP | BP | CP are the closest path together.

belong to which category ?